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Paleomagnetic and Geologic Data Indicating 2500 km of Northward Displacement for the Salinian and Related Terranes, California

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Results of a pair of paleomagnetic studies of sedimentary rocks in the Salinian terrane of westernmost California indicate a northward displacement of about 2500 km since Cretaceous time. Stratigraphic relations suggest that the Salinian terrane became amalgamated with several others by the Late Cretaceous and all underwent the same northward translation. This composite terrane (allochthon) became sutured to cratonal North America by early Tertiary time. One paleomagnetic study involved Upper Cretaceous turbidites of the Pigeon Point Formation. Fine-grained, thin-bedded turbidites yielded the most reliable results, whereas medium- to coarse-grained sandstones are not always stably magnetized. Contorted layers resulting from soft sediment deformation usually gave directional results parallel to those of undeformed sedimentary layers, thereby implying a postdepositional remanence process. There is no indication of inclination error or other processes affecting the fidelity of the remanent magnetization. Paleomagnetic results from four separate locations in the Pigeon Point Formation gave positive fold tests and mean inclination values near 36°. A mean geomagnetic latitude calculated from these results is 21.2° ± 5.3°, much more southerly than the value of 46.5° ± 2.3° calculated for Pigeon Point using paleomagnetic pole data from Late Cretaceous rocks from cratonic North America. Declination values are not antipodal for normal and reversed polarity sites. Analysis of depositional current directions from sandstone beds of the same outcrops shows that the difference is the result of Neogene rotations of small crustal blocks adjacent to the San Gregorio fault. In a sequence of Paleocene turbidites that crop out at Point San Pedro, four sample localities yielded positive fold tests with mean inclination values near −40°. Declination values, however, were spread over a range between 230° and 320°. Comparison with depositional current directions from the same outcrops indicates clockwise rotations related to Neogene right slip on the Pilarcitos fault. The mean geomagnetic latitude calculated for the Point San Pedro sediments is 24.5° ± 4.0° and is more southerly than the 43.3° ± 2.5° latitude expected for that location. For the Salinian terrane and the west coast of cratonic North America, absolute rates of latitudinal motion were 8.5 cm/yr northward and 3 cm/yr southward, respectively.

INTRODUCTION

Upper Mesozoic and lower Cenozoic rocks in California can be conveniently lumped into three petrogenetic groups: (1) intrusive and extrusive crystalline and associated high-temperature metamorphic rocks of sialic arc sequences, for example, the Sierra Nevada batholith, (2) arkosic sediments of forearc sequences, for example, the Great Valley sequence, and (3) a heterogeneous assemblage of graywacke, limestone, chert, greenstone, and high-pressure metamorphic rocks of subduction complexes, for example, the Franciscan assemblage. This triad represents a type example for the arc-trench-gap model of *Dickinson* [1970]. In map pattern, rocks of these three categories fall generally into three parallel, northwest trending belts.

The one obvious departure from this geometric pattern is the apparent doubling up of two such triad sequences in central California. The long-standing explanation involves right-lateral displacement of the San Andreas fault system [*Hill and Dibblee*, 1953; *Page*, 1981]. A minimum slip of 520 km is required to adjoin palinspastically granite of the Salinian terrane cropping out along the coast of northern California with inferred autochthonous granitic remnants of the Sierran trend in southern California. This magnitude of slip is well within the 600- to 1000-km limits of Neogene displacement between the Pacific and North American plates [*Atwater*, 1970].

The San Andreas fault alone can account for no more than 315 km of post-Eocene slip [*Nilsen and Clarke*, 1975]; this limit has generated the notion of a proto-San Andreas fault. In southern California, however, Precambrian strata are displaced no more than 305 km [*Crowell*, 1962], so the validity of the proto-San Andreas model becomes suspect. *Graham and Dickinson* [1978] cleverly eliminate the need for much of the slip on a proto-San Andreas by providing evidence for as much as 115 km of displacement on the San Gregorio–Hosgri fault trend. This more northerly trending fault joins the San Andreas fault from the west at a small angle, north of San Francisco. Palinspastic restorations of such large magnitudes for the southern portion of the San Gregorio–Hosgri fault trend remain enigmatic, as this fault system cannot be traced in the subsurface offshore, south of the Transverse Ranges trend. Regardless, at least 100 km of slip is still unaccounted in order to align granitic rocks of the Salinian terrane as a southward extension of the Sierra Nevada batholith. Because of the lack of any supporting evidence for a proto-San Andreas fault in southern California the validity of the Salinian-Sierran tie remains in question.

Additional geologic considerations have been raised from time to time that also mar the tidy concept indicating offset of a simple arc-trench-gap configuration. *Gastil* [1975] noted that the rocks in the Salinian block, if correlated with the Sierra Nevada–Peninsular Ranges batholithic terranes, are representative of only the eastern lithofacies of these larger terranes.

This prompted *Page* [1981] to suggest that a region west of the Salinian terrane, equivalent to the entire Sierran foothill region, had been lost owing to tectonic erosion processes. *Silver* [1982], however, offers a tectonic model that involves early Tertiary west directed thrusting to transport the eastern lithofacies into an allochthonous western region. Neogene strike-slip faulting has subsequently dispersed these strata along the west margin of California. Alternatively, *Dickinson* [1983] proposes 560 km of left slip on the Naciemento fault

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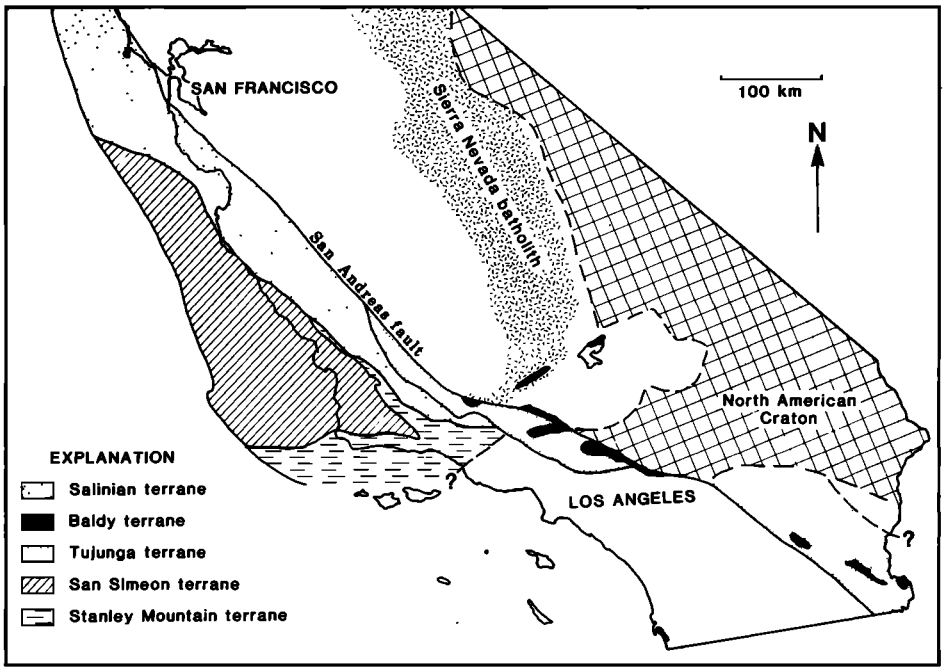


Fig. 1. Map of part of California showing present distribution of tectonostratigraphic terranes composing the Santa Lucia–Orocopia allochthon. For distribution and character of other terranes in California, see *Blake et al.* [1982]. Neogene to recent right slip of the San Andreas fault system is disrupting the terrane amalgam.

prior to Tertiary faulting on the San Andreas fault system. Palinspastic restoration for such left-lateral movement restores the Salinian terrane to an eastern batholitic position; however, this south directed motion must be younger than 80-m.y.-old Salinian plutonism, and latest Cretaceous south directed slip is opposite to that inferred from our paleomagnetic experiments.

These models involving lost or westerly displaced parts of a Sierran arc segment or hypothetical episodes of major left slip assume authenticity of the correlation between Salinian and Sierran rocks. This premise has been questioned owing to differences in mineralization and prebatholithic stratigraphy [*Ross*, 1977, 1978]. Thus, during the past three decades much has been written about paleotectonic and paleogeographic reconstructions of California. Even though in many aspects geologic and plate tectonic data are mutually supportive, no model unequivocally explains the geologic history of the Salinian terrane wholly in the context of now contiguous Californian terranes.

Many paleomagnetic studies from the cordillera of western North America consistently show discordance indicating clockwise rotations and northward translations relative to stable cratonic North America [*Beck*, 1976, 1980]. Differences in the paleomagnetic data for separate areas of the cordillera reflect differences in their tectonic history. For instance, paleomagnetic results from the Tertiary of Oregon and Washington suggest rotations but no translations of crustal blocks possibly owing to nearby oblique underthrusting. Jurassic plutons of the Sierra Nevada, on the other hand, have paleomagnetic results nearly consistent with those of stable North America, allowing at most a small counterclockwise rotation of one isolated pluton [*Gromme et al.*, 1967]. Plutons of the southern California batholith have moved northward about 12° since their Early Cretaceous emplacement ages [*Teissere and Beck*, 1973]. Miocene volcanic and selected sedimentary strata in southern California [*Kamerling and Luyendyk*, 1979; *Luyendyk et al.*, 1980; *Hornafius et al.*, 1981] indicate coastal blocks with up to 110° of clockwise rotation and possibly as much as 10° of northward translation; consistent with Neogene dextral shear along the western edge of North America [*Beck*, 1980]. Paleomagnetic data from blocks within Franciscan terranes indicate large-scale northward displacements, possibly from the southern hemisphere [*Alvarez et al.*, 1980]. Paleomagnetic study of the rocks of the Salinian terrane offers the opportunity of discerning the original position of this terrane relative to the Sierra Nevada granitic rocks and of evaluating possible correlations.

It was with this understanding that we designed a paleomagnetic experiment in order to acquire independent data regarding paleolatitudinal positions of this anomalous granitic terrane. The results are outlined in this paper; they clearly show why past efforts to link Upper Jurassic and Cretaceous terranes east and west of the San Andreas fault remain equivocal.

TERRANES

The paleomagnetic experiment discussed in this paper is focused on rocks of the Salinian terrane, yet on the basis of stratigraphic analyses [*Vedder et al.*, 1983] and other paleomagnetic experiments [*McWilliams and Howell*, 1982] we can infer paleolatitudinal shifts of adjoining terranes. Figure 1 shows the present locations of the disparate terranes that presumably share a similar displacement history to the Salinian block since latest Cretaceous time [*Blake et al.*, 1982]. A brief and previously unpublished description of the geologic/stratigraphic character of the Salinian and adjoining terranes is provided below.

San Simeon/Stanley Mountain Terranes

The San Simeon and Stanley Mountain terranes are lithologic correlatives of the Franciscan assemblage and Great

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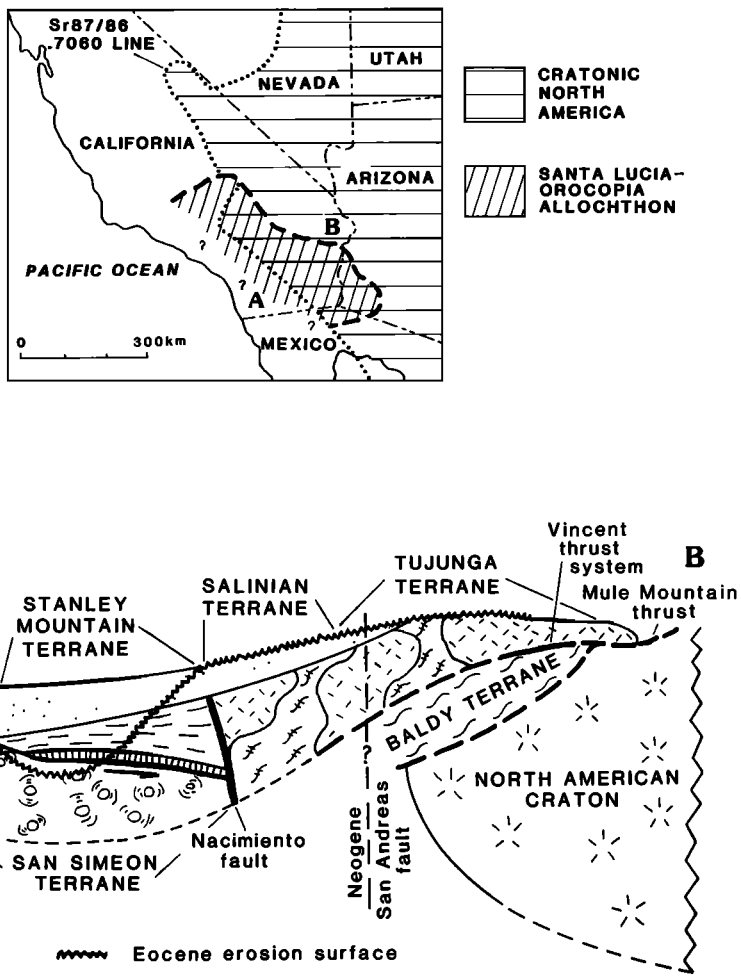


Fig. 2. Generalized location and cross section of the Santa Lucia–Orocopia allochthon for Eocene time immediately following accretion into the North American Craton.

Valley sequence, respectively. The San Simeon terrane is largely a melange with lithofacies of graywacke, serpentine, greenstone, chert, and locally blueschist knockers. Cropping out along the coast are a number of coherent, though isoclinally folded, Upper Cretaceous graywacke subterranes, such as the Cambria slab [*Hsu*, 1969; *Howell et al.*, 1977]. Petrographic comparisons show similar compositions among sandstones from the graywacke subterranes, Stanley Mountain and Salinian terranes [*McClure*, 1969; *Lee-Wong and Howell*, 1977]. The Stanley Mountain terrane structurally overlies the San Simeon terrane and is composed of a mid-Jurassic ophiolite overlain by 50–100 m of Upper Jurassic tuffaceous chert which grades upward into 1000–4000 m of Upper Jurassic and Cretaceous arkosic turbidites. Uppermost Cretaceous strata, composed locally of granite grus, were deposited in a fluviodeltaic and marine environment. A contiguous Salinian terrane is the suggested source for this granitic debris [*Howell et al.*, 1977; *Vedder et al.*, 1983].

Lithologic and paleontologic studies involving the Stanley Mountain terrane indicate remarkable similarities between this terrane and a seemingly equivalent terrane east of the San Andreas fault (i.e., the Del Puerto terrane of *Blake et al.* [1982]). Nonetheless, preliminary paleomagnetic data by us and by *McWilliams and Howell* [1982] suggest that the San Simeon and Stanley Mountain terranes are exotic to California.

Salinian Terrane

The Salinian terrane is composed of granitic batholithic rocks, with 80- to 110-m.y.-old crystallization ages [*Mattinson*, 1978; *Ehlig and Joseph*, 1977], that intrude the Mesozoic or older Sur-Series Metamorphic Complex of *Trask* [1926]. Depositionally lapping across these basement sequences are Upper Cretaceous strata characterized by nonmarine to deep marine deposition in basins affected concurrently by tectonism. Petrographic analyses of the sandstones indicate local Salinian provenances, though locally voluminous conglomerates composed of quartzite and siliceous volcanic clasts require other cordilleran source areas as well [*Howell et al.*, 1977].

Tujunga Terrane

The Tujunga terrane is a composite of the Precambrian San Gabriel and Joshua Tree terranes of *Powell* [1981] intruded by Jurassic and Cretaceous granitic rocks. Upper Cretaceous shallow marine lithofacies that lap onto the La Panza Granite of the Salinian terrane continue southeastward and also lie depositionally on gneissic rocks of the Tujunga terrane at Barrett Ridge. Locally, in the San Gabriel Mountains, Precambrian crystalline basement rocks are overlain by Upper Cretaceous and Paleocene rocks of the San Francisquito Formation [*Kooser*, 1980]. We believe these Upper Cretaceous strata are correlative lithofacies. Moreover, rubidium-strontium iso-

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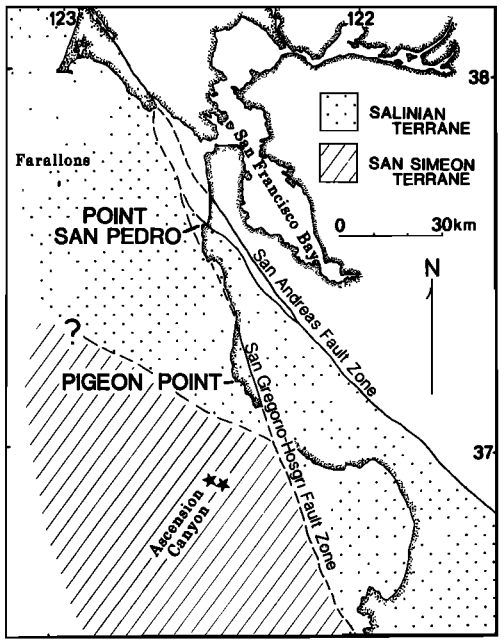


Fig. 3. Location map showing paleomagnetic collecting sites, major faults, and terrane boundaries of central coastal California. Stars near Ascension canyon mark positions of dredged samples of serpentine and graywacke from Franciscan basement.

topic and age data as well as a distinctive polka-dot texture in some of the plutonic bodies argue strongly for a genetic link between quartz monzonite plutons of the Salinian terrane and plutons in the Tujunga terrane [*Ehlig and Joseph*, 1977; *Smith*, 1977; *Joseph and Ehlig* 1981]. These geologic relations indicate that by latest Cretaceous time the disparate basement terranes of the Salinian and Tujunga terranes must have formed a single composite basement complex.

The crystalline rocks of the Tujunga terrane are everywhere the hanging wall of the Vincent-Orocopia-Chocolate Mountain thrust system [*Crowell*, 1981]. This structural position and the anomalous character of the Precambrian rocks suggests that the Tujunga terrane is allochthonous and is possibly exotic to the cratonal framework of western North America [*Ross*, 1977; *Haxel and Dillon*, 1978; *Silver*, 1971].

Baldy Terrane

Structurally beneath the Tujunga terrane and cropping out as fensters is the schistose tectonite of the Baldy terrane [*Haxel and Dillon*, 1978]. These rocks are mapped locally as the Pelona, Orocopia, and Rand schists. The crossite-bearing schist bodies that formed under conditions of high temperature and pressure are capped by mylonite of the Vincent-Orocopia–Chocolate Mountain thrust system. The base of this metamorphic terrane has never been observed. The protoliths of these rocks are mostly graywacke with lesser amounts of chert, limestone, and basalt. The age of deposition of the protoliths is unknown, whereas the age of the latest metamorphic event is constrained by K-Ar dating at latest Cretaceous to earliest Tertiary time [*Ehlig et al.*, 1975].

Terrane Amalgamation

This brief description of the geologic terranes depicted in Figure 1 points out a variety of geologic criteria that suggest the San Simeon, Stanley Mountain, Salinian, and Tujunga terranes were spatially linked by Campanian time (about 75–80 m.y. B.P.). Spatial relations between these amalgamated terranes and the Baldy terrane prior to faulting on the Vincent thrust are not known. The present distribution of these terranes as extended northwest trending crustal slivers is largely due to tectonic dispersion along the Neogene San Andreas fault system. Figure 2 is a small-scale palinspastic reconstruction depicting a generalized configuration for this pre-Neogene agglomeration of terranes, hereinafter called the Santa Lucia–Orocopia allochthon [*Vedder et al.*, 1983]. A schematic cross section of this conceptual allochthon illustrates the structural relations among the component terranes.

PALEOMAGNETIC EXPERIMENTS

Our tectonic reconstruction for the terranes west of the San Andreas fault in central California departs from earlier models owing to insights gained from the paleomagnetic experiments. The validity of these models is wholly dependent upon the accuracy of the paleomagnetic results. The following sections describe the details of two paleomagnetic experiments, using sedimentary rocks of the Cretaceous Pigeon Point Formation and Paleocene strata at Point San Pedro.

Geology of Pigeon Point

The Cretaceous Pigeon Point Formation [*Hall et al.*, 1959] crops out along a 16-km section of the coastline of central California (Figures 3 and 4). Fossils within the section are rare, but all indicate a Campanian to Maestrichtian age (78–65 m.y.) for the section [*Hall et al.*, 1959]. The stratigraphy and sedimentology of the sequence indicate a progradational deposition sequence with submarine fan to shelf environments [*Howell and Vedder*, 1978]. Thick-bedded, coarse-

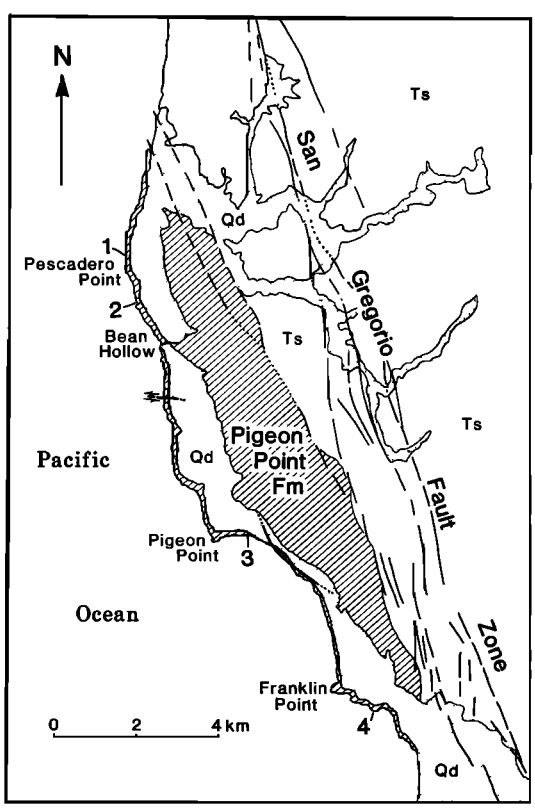


Fig. 4. General geology of the Pigeon Point area, showing paleomagnetic sampling localities, major fault strands, and geologic units. Symbol Ts denotes Tertiary sedimentary rocks and Qd refers to various Quaternary deposits. Site numbers correspond to Table 1.

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to medium-grained turbidites dominate the lower portion of the formation. In the middle part, within upward thinning stratigraphic intervals, are many fine-grained layers of silt and mud. At the top of the section, conglomerate and pebbly mudstone are overlain by hummocky cross-stratified strata of Bolsa Point or the downflow equivalent thick-bedded turbidites and grain flow deposits of Franklin Point.

The basement rock underlying the Pigeon Point Formation is not known with certainty. Immediately west and northwest of the area the Pacific continental shelf is relatively broad and featureless. Granitic rocks crop out on the Farallon Islands and compose the northern end of the Farallon structural high. Geophysical evidence indicates that this high may continue southeasterly as the Pigeon Point structural high [*McCulloch et al.*, 1980]. These relations suggest that the Pigeon Point Formation is part of a faulted segment of the larger Salinian granitic terrane [*Howell and Vedder*, 1978]. Southwest of Pigeon Point, in the headward part of Ascension Canyon, dredged samples of serpentine and graywacke indicate nearby Franciscan basement [*Mullins and Nagel*, 1981]. They infer from these relations that an offset extension of the Sur-Nacimiento fault system lies along the south limit of the featureless shelf southwest of Pigeon Point and just north of the heads of Pioneer and Ascension canyons which are cut into the shelf (Figure 3). This inference limits the right slip on the San Gregorio–Hosgri fault system to less than 90 km and suggests that the Pigeon Point Formation is a bona fide part of the Salinian terrane.

The entire sequence has been structurally deformed in three distinct episodes [*Joyce*, 1981]. Folding around horizontal northwest trending axes occurred before Oligocene rocks of the Vaqueros Formation were unconformably deposited on the Pigeon Point strata. Additional faulting and folding occurred in the Neogene, probably associated with the nearby San Gregorio fault. The most recent deformation involves tilting of older fault planes and Pleistocene marine terraces [*Weber and Lajoie*, 1979]. This deformation also reflects slip on the San Gregorio fault.

Paleomagnetic Procedures

Core samples for paleomagnetic analysis were taken using a portable gasoline-powered drill and were oriented at the outcrop with a magnetic compass. Fine-grained, hemipelagic mud layers in the upper part of turbidites (Bouma T_e intervals) [*Bouma*, 1962] were preferentially sampled, though occasional coarser-grained horizons were also drilled. This was done in part on the assumption that coarse-grained samples tend to have larger proportions of viscous remanent magnetization (VRM) within their natural remanent magnetization (NRM) and are thus poorly suited to magnetic cleaning. The assumption proved to be correct, as few very coarse grained samples could be completely magnetically cleaned.

Coarse-grained layers were avoided also because they frequently showed iron oxide staining. The strata directly beneath the unconformity at the base of the overlying Pleistocene marine terrace deposits typically show a strong orange-red color from oxidation. This color penetrates deeply into coarse-grained layers within the Pigeon Point Formation, sometimes to sea level. By avoiding the coarse-grained layers we could avoid the overprinting magnetizations of the authigenic iron oxide minerals which usually obscure the original magnetic direction of a sample. Bedding attitudes were measured at frequent intervals during the sampling in order to correct properly for tectonic tilts of up to 88°.

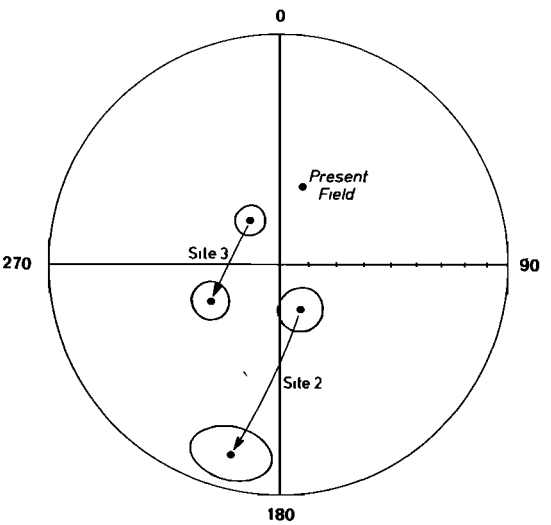


Fig. 5. Equal-area diagram of magnetization directions uncorrected for bedding attitudes with 95% confidence circles from sites 2 and 3 before and after af cleaning showing removal of present field component.

Samples were cut in the laboratory into 2.2-cm length cores. The direction and intensity of NRM was measured in one specimen from each core. The intensity of NRM was quite low, usually between 8×10^{-7} and 3×10^{-6} emu/cm^3. The weakness of magnetization dictated caution in the magnetic cleaning of the samples. Each specimen was progressively demagnetized by 25- or 50-Oe steps in an alternating magnetic field (af) to at least 400-Oe peak field strength. The af demagnetization removed overprinting components of secondary magnetization from most specimens and isolated their primary magnetization. Trial thermal demagnetizations were performed on second specimens from cores for which af cleaning data already existed. The results showed that although thermal cleaning occasionally can remove secondary components of magnetization as well as af cleaning, it usually does not work as well for these rocks.

The principal component of secondary magnetization was VRM which tended to bias each specimen's NRM direction toward the present magnetic field before the tectonic correction was applied (Figure 5). Coarser-grained specimens possessed relatively larger VRM components and occasionally could not be completely cleaned. We ascertained the tilt uncorrected direction of the primary magnetization in each sample with equal-area projections of the directions and vector component diagrams of the progressive demagnetization data. Demagnetizer noise routinely became apparent and troublesome at peak af strengths greater than 200 Oe.

After af cleaning, each sample was judged whether it was completely free of secondary magnetic components. Of the original 154 samples taken in the field, 36 (23%) were found to be incompletely cleaned and were rejected from further analysis. The three reasons for rejection were (1) lack of a stable cleaned direction, (2) failure to remove present field direction from samples, and (3) random directions in sediment layers which had undergone soft sediment disruption. Coarse-grained samples mostly failed to show stable cleaned directions (28 of 36) and instead have low coercivities resulting in erratic directions of magnetization due to the noise of demagnetization (Figure 6*a*). Removal of secondary magnetic components is usually impossible for samples visibly tinted with authigenic ferric oxide minerals related to the presence of the

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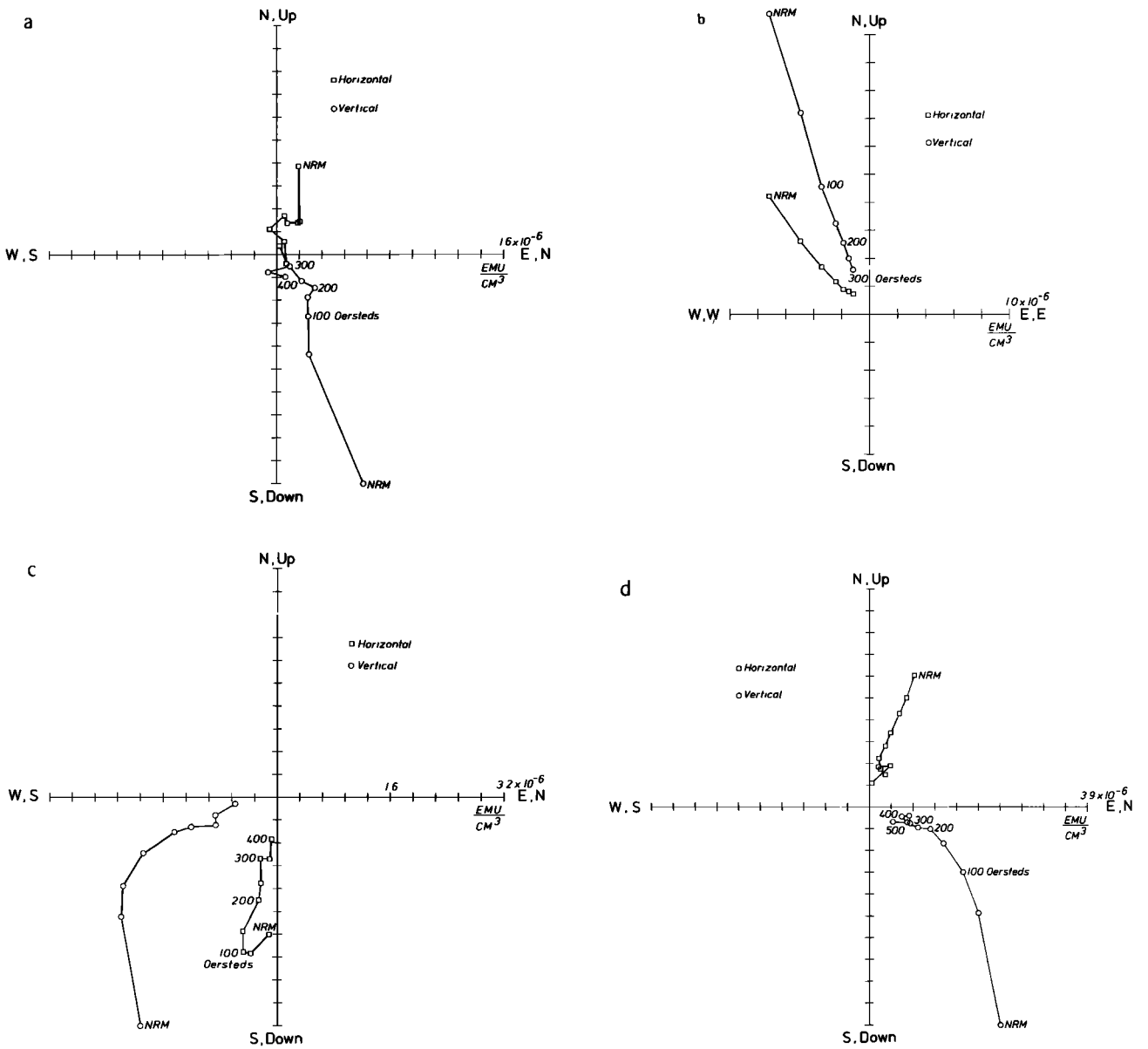


Fig. 6. Orthogonal vector component diagrams of alternating field demagnetization of samples of the Pigeon Point Formation. (*a*) Sample from site 1 showing erratic cleaning behavior due to coarse grain size. (*b*) Sample from site 2 showing poor cleaning due to authigenic ferric oxide overprint. (*c*) Sample from site 1 showing cleaning producing linear descent to the origin. (*d*) Sample from site 4 showing good magnetic cleaning and a stable direction of magnetization but no perfectly linear segments.

overlying Pleistocene terrace deposits (11 of 36); indeed, at least five samples were intentionally collected despite evidence of oxidation to verify our a priori avoidance of weathered outcrops. The vector component diagrams for these types of samples are characterized by a stronger intensity, higher coercivity, single-component magnetization having an uncorrected direction close to the present field. Although layers characterized by soft sediment disruption usually gave cleaned magnetic results equivalent to undisturbed layers, in at least four samples they did not, for reasons described in the following section.

The 118 samples cleaned of secondary magnetic components can be classified in two basic categories from vector component diagrams. Sixty one of the samples have diagrams with linear segments that pass through the origin for some portion of their demagnetization behavior (Figure 6*c*). The remaining 57 samples accepted show good magnetic cleaning and stable magnetic directions but do not produce perfectly linear segments on vector component diagrams (Figure 6*d*).

The expected paleomagnetic directions and paleolatitudes at the sampling sites for Cretaceous and Tertiary time (Table 3) were calculated from a selected set of paleomagnetic poles derived from rocks in cratonic North America [*Irving and Irving*, 1982; *Diehl et al.*, 1983].

Nature and Timing of Magnetization

Consideration of the timing and process of acquisition of the primary remanent magnetization is especially important in this study because in these sedimentary rocks such factors are less well understood than, for example, in lavas and also because the conclusions we will draw depend importantly on whether or not the inclination of the primary magnetization is

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an unbiased record of the geomagnetic inclination at the time the rocks were formed. There is extensive literature (reviewed by *Verosub* [1977]) on the mechanisms by which a sediment or sedimentary rock acquires magnetization, and these papers include a number of experimental studies as well as investigations of NRM. A problem that is frequently discussed is the inclination error, which is the tendency for the detrital remanent magnetization (DRM) acquired by a sedimentary during deposition to have an inclination shallower than that of the applied field. All of the theoretical analyses and most of the experimental studies dealing with inclination error have assumed or created a situation in which the magnetization is entirely acquired as the individual grains come to rest at the sediment-water interface. Under such conditions, inclination error is both predicted and observed. What is crucial to the present study, however, is the likelihood that the NRM in the sedimentary rocks we have studied was acquired at times shortly after deposition. Under such a circumstance the physical environment of the magnetic grains as they become locked in place would be much more nearly isotropic than at the sediment-water interface, and therefore systematic inclination error would be less likely to occur [*Verosub*, 1977]. Several natural and experimental studies bear this out. The first comprehensive investigation was that of *Opdyke and Henry* [1969], who analyzed paleomagnetic results from a large number of deep-sea cores, all younger than 2 m.y. and widely distributed in latitude. The measured inclinations conformed closely to the field of a geocentric axial dipole. The postdepositional magnetization resulting from redeposition of slurries of deep-sea sediment was investigated by *Kent* [1973], who found no systematic bias of inclination for a wide range of experimental conditions. Kent also points out the importance of bioturbation as a postdepositional mechanism for NRM acquisition in deep-sea sediments; this process can evidently be effective to depths as great as 60 cm [*Verosub*, 1977]. A useful comparison of NRM in marine sedimentary rocks and associated basaltic lavas was published by *Simpson and Cox*, [1977]. Investigating Eocene rocks of western Oregon, they found virtually identical inclinations in the sedimentary Tyee and Flournoy formations, in the underlying Siletz River Volcanic Series, and in the overlying Yachats basalt.

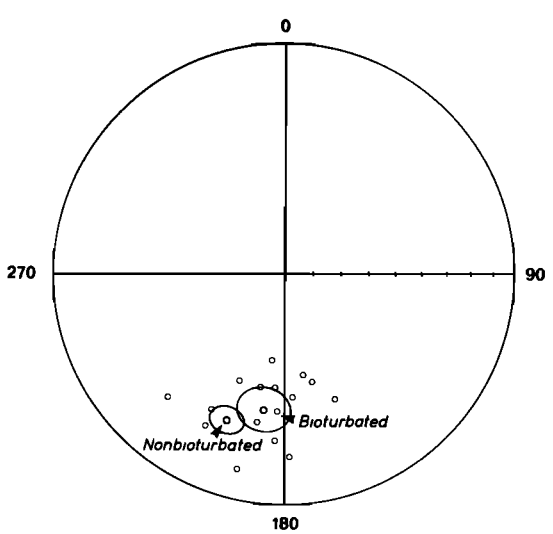


Fig. 7. Equal-area diagram of all directions, mean direction, and 95% confidence circle for bioturbated samples compared to mean direction and 95% confidence circle for nonbioturbated samples from site 1. Data corrected for dip and strike.

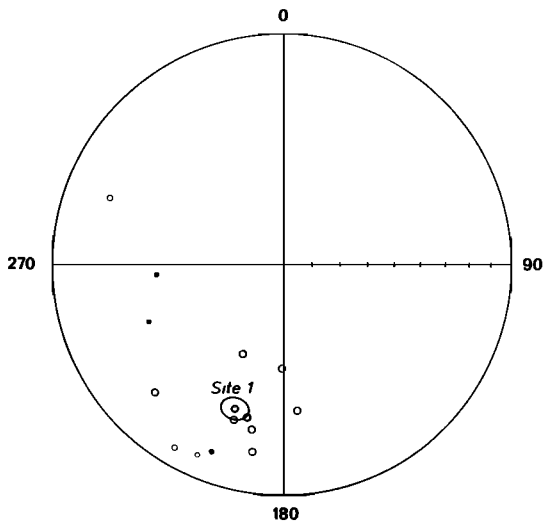


Fig. 8. Equal-area diagram of directions for 14 samples chaotically disturbed by soft sediment deformation compared to the mean direction and 95% confidence circle for all samples from site 1. Large open symbols represent data included in the mean direction for site 1, smaller open and solid symbols represent data which was rejected from mean. Open symbols represent data on upper hemisphere; solid symbols on lower hemisphere. Data corrected for dip and strike.

Verosub [1977] also discussed the process of postdepositional detrital remanent magnetization (PDRM) at some length, emphasizing that, in general, no inclination error should be expected and also emphasizing the importance of critical water content in the magnetization process. When, after deposition, the amount of pore water in the sediment decreases below the critical value, the magnetic grains are no longer free to rotate in response to mechanical influences, and their statistical alignment with the geomagnetic field becomes parmanent. Results of experiments in which the growth of PDRM was measured during compression and dewatering of artificial and natural sediments have recently been reported by *Hamano* [1980]. In these experiments the water content was expressed as the void ratio of the sediment, and the critical void ratios were found to be 1.4 for a synthetic sediment and 6.5 for a natural sediment. No convincing evidence for inclination error was found, although the results were somewhat scattered.

Denham and Chave [1982] have recently shown that by assuming an exponential increase in viscosity (as experienced by the magnetic particles) with increasing depth-time in the consolidating sediment, most of the magnetization should be acquired in a relatively narrow zone of depth-time. They also point out that the relative viscosity is empirically well known to be an exponential function of the relative particle concentration. Although actual density profiles are complex, the preliminary analysis by Denham and Chave provides some support for the concept of the lock-in zone, that is, the depth range corresponding to the range of critical water content.

Bioturbated horizons are a common occurrence within the finer-grained parts of the Pigeon Point Formation. Evidence of bioturbation consists of Ophiomorpha-like burrows up to 1 cm in diameter that cross cut bedding surfaces, Condrites-like forms consisting of 1-mm-wide tubes that are irregularly distributed within fine-grained Bouma T_e layers [*Bouma*, 1962],

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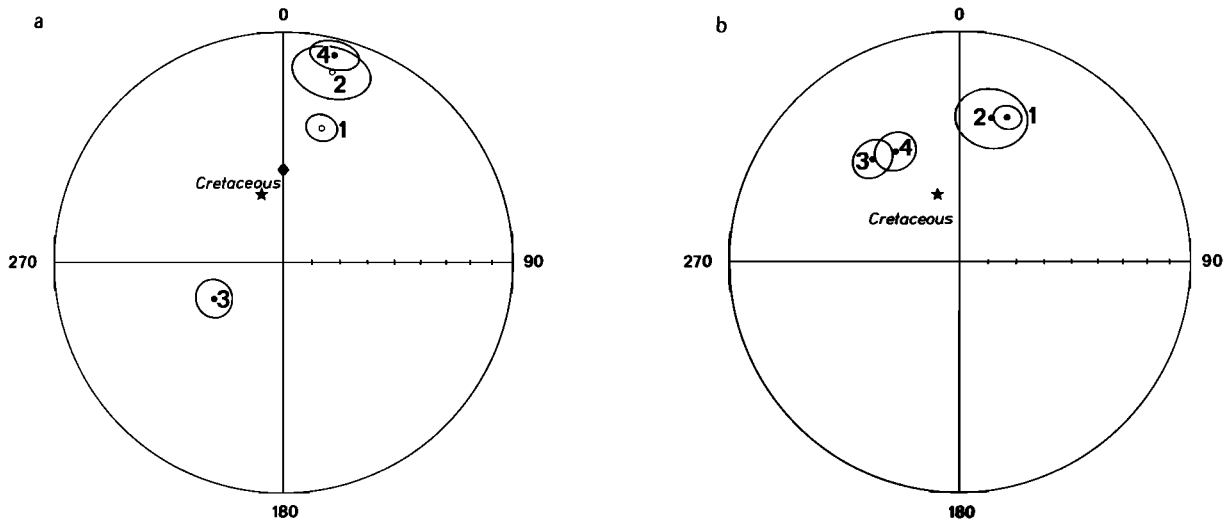


Fig. 9. Equal-area plots of (*a*) uncorrected and (*b*) structurally corrected field directions from the upper Cretaceous Pigeon Point Formation. Numbers indicate the means and 95% confidence circles for results from 1, Pescadero Point; 2, Pebble Beach; 3, Pigeon Point; and 4, Franklin Point. The reversed field directions at Pescadero Point and Pebble Beach have been converted to south seeking directions. The star denotes the expected Upper Cretaceous field direction for normal polarity at the position of Pigeon Point if it were part of stable North America. The diamond shows the present axial dipole direction for the collecting sites.

and zones of mixed sediment displaying a mottled appearance. Samples were taken from layers showing all forms of bioturbation. A moderate difference can be seen between the mean direction of the bioturbated samples ($n = 16$) and that of the nonbioturbated samples ($n = 47$) (Figure 7). If mean inclinations only are compared, the two populations are not significantly different at the 95% confidence level [*McFadden and Jones*, 1981]. This indicates that either no bias of the DRM has occurred within the Pigeon Point strata or that a PDRM has been created within the entire section, bioturbated or not. The difference between the two data distributions is no greater than between different areas within single sites containing both bioturbated and nonbioturbated samples.

Soft sediment deformational structures occur at many areas within the Pigeon Point sequence and have been extensively sampled at site 1 for information on the timing of the magnetization. Flaps of sediment, flame structures, and small folds reflect disruption, up to 1 m deep in underlying sediment, resulting from mass-flow deposition in high-energy environments. We sampled these structures to see if soft sediment deformation, which is manifested by both chaotic and coherent folding, also disrupted the magnetization within the affected sediments.

Fourteen samples were taken from beds in site 1 that are chaotically deformed (Figure 8), and of these, eight had magnetizations sufficiently close to the site mean direction from undeformed beds that the site mean direction was recalculated to include them. Six samples were not included in the site mean direction though their directions of magnetization were not randomly distributed but instead were loosely grouped in the same quadrant of the equal-area net containing the site 1 mean direction. This suggested to us that at the time of disruption of these beds the DRM had been at most only partially set.

In two instances we were able to sample beds which were coherently deformed by soft sediment processes. By measuring fold axes of soft sediment deformations as well as structural tilt it is possible to perform a complete structural correction for these two samples. One sample, when first untilted and then unfolded about the inclined soft sediment deformation axis gives a direction of magnetization unrelated to the characteristic direction for site 1 and probably is still contaminated with magnetization acquired in the present field. The other sample produces an inclination value near $-40°$ and close to the characteristic direction for site 1 after complete structural correction. Thus at the time of soft sediment deformation this specimen was completely magnetized, and the delay in timing of its magnetization no greater than the time to deposit about 1 m of overlying sediment, probably less than 1000 years. This estimate is derived from the conclusion of *Howell et al.* [1977] that the 3000 m of strata accumulated during an approximate 3-m.y. time interval.

Results of the sampling from beds showing bioturbation and soft sediment deformation convince us that the magnetization of the Pigeon Point Formation was acquired soon after deposition. Assuming sediment accumulation rates of 1 m per 1000 years, combined with the fact that disrupted zones up to 1–1.5 m thick are partially magnetized before disruption, the PDRM is probably acquired within 10^4 years. Because the

TABLE 1. Paleomagnetic Directions and Related Dispersion Parameters After af Demagnetization, Pigeon Point Formation

Site	Data	*N*	*R*	*k*	α(95)	*I*	*D*
1	uncorr	63	58.79874	14.8	4.8	39.2	195.9
	corr		59.14859	16.1	4.6	−34.4	198.2
2	uncorr	12	11.25132	14.7	11.7	16.5	193.5
	corr		11.25269	14.7	11.7	−35.5	191.7
3	uncorr	27	25.53619	17.8	6.8	62.4	243.1
	corr		25.31622	15.4	7.3	40.7	320.1
4	uncorr	16	15.50722	30.4	6.8	7.9	13.9
	corr		15.46725	28.2	7.1	43.9	329.5

Two data sets, one structurally corrected (corr) and the other uncorrected (uncorr) are given for each of four outcrop areas. *N*, number of independently oriented cores; *R*, length of the resultant vector; *k*, Fisherian dispersion parameter; α(95), radius of the cone of 95% confidence about the mean direction; *I*, inclination in degrees downward; *D*, declination in degrees east of north. Site numbers correspond to those in captions of Figures 4 and 8.

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magnetization of the Pigeon Point Formation is postdepositional, the mechanisms that can cause inclination error are inapplicable in this case. If there was a biased DRM at first, the postdepositional period has allowed sufficient time to remove the bias.

Paleomagnetic Results From Pigeon Point Formation

Paleomagnetic collections were made at four locations in the Pigeon Point Formation (Figure 4): (1) Pescadero Point, (2) Pebble Beach, (3) Pigeon Point, and (4) Franklin Point. The magnetic directions for each sample from each location were corrected for tectonic tilt by rotation around the strike, and then all sample directions at a site were averaged (Figure 9 and Table 1). Reversed magnetic polarity was found in strata at Pescadero Point and Pebble Beach and normal polarity in strata at Pigeon Point and Franklin Point. (In the following, the reversed data are converted to south seeking directions.) Uncorrected mean magnetic directions for the four sites show considerable scatter (the precision parameter $k = 1.7$ for $n = 4$ sites), particularly in inclination. After correction for dip of bedding this scatter diminishes ($k = 12.4$), and inclinations become consistent near 37°.

Positive fold tests also exist for pairs of locations having the same polarity. Mean magnetic directions for locations 1 and 2 move considerable angular distances upon unfolding and become indistinguishable at the 95% confidence level [*McFadden and Jones*, 1981]. Directions at locations 3 and 4, which have very dissimilar structural attitudes, move even greater angular distances upon structural correction, and also become indistinguishable at the 95% confidence level [*McFadden and Jones*, 1981]. Declination values for the normal magnetic directions are mutually compatible as are the declinations for the reversed magnetic directions. The folding which has affected these four locations occurred before the strata of the Pigeon Point Formation were deformed and eroded and before deposition of the overlying early Miocene (30 m.y.) shallow marine sandstone. The fold test thus demonstrates that the magnetization has been stable at least since early Miocene time.

Although the fold tests give positive results, the declinations of the two polarity groups differ from each other by 50°. The fact that the inclinations are all similar indicates that simple rotation of bedding around the strike is an incomplete structural correction. One interpretation could be that the fold axes are not horizontal but instead have significant plunges. The

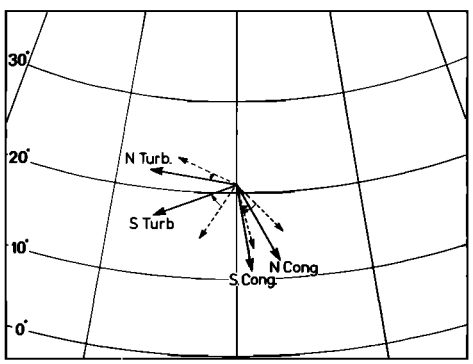


Fig. 10. Current directions from Pigeon Point Formation as measured (dashed) and after normalization to 0° declination (solid) at their probable latitude of origin in Late Cretaceous time. Turb represents paleocurrent data from turbiditic sands and Cong. from clast imbrication.

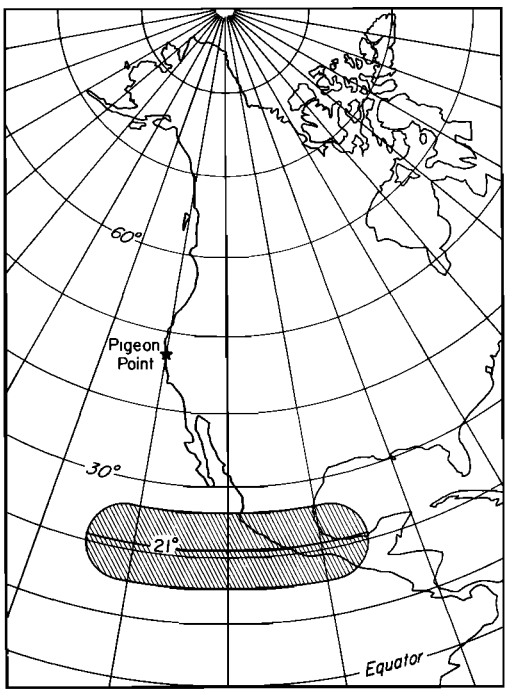


Fig. 11. Equal-area map of North America showing suggested latitude band of origin for Cretaceous Pigeon Point Formation. Latitude-longitude grid corresponds to interpolated Cretaceous-Paleocene paleomagnetic pole position (see Table 3).

structure of the Pigeon Point Formation cannot be described as simple or cylindroidal folding; at each of the sites the dips are nearly homoclinal, while the bedding planes of all four sites do not have a common intersection. Rather, the structure appears to be a series of fault-bounded blocks. The reversely magnetized locations are spatially separated from the normally magnetized locations by 10 km along the coastline, and they are on opposite sides of a major east-west trending fault located 2 km south of Bean Hollow [*Howell and Joyce*, 1981]. Howell and Joyce describe this fault as having at least 800 m of vertical throw based on direct stratigraphic separation and an unknown amount of horizontal slip. The differences in declination could then have resulted from significant relative rotations of the structural blocks north and south of this and possibly other faults. As the present outcrop area of the Pigeon Point Formation lies just west of the San Gregorio fault zone (Figure 4), it is easy to imagine how these rotations might occur. Because we surmise from the granitic sedimentary content of the Pigeon Point Formation [*Lee-Wong and Howell*, 1977] that the site of its deposition was contiguous to the cordillera of North or Central America, it follows that the probable path of motion was roughly coincident with the Late Cretaceous paleomeridian. The observed declination for the Pigeon Point Formation must at least partly reflect the orientation of the coastline where it originated and will also have been affected by the relative internal rotations associated with the northward displacement.

Additional support for block rotations as the explanation of the declination differences can be derived from an analysis of the paleocurrent data available for the Pigeon Point Formation [*Howell and Joyce*, 1981]. Paleocurrent directions for turbidites in the northern block give 295° azimuths, while in the southern block, equivalent indicators give 215° azimuths. Similarly, clast imbrication data from submarine canyon conglomeratic beds in the northern block give 165° azimuths,

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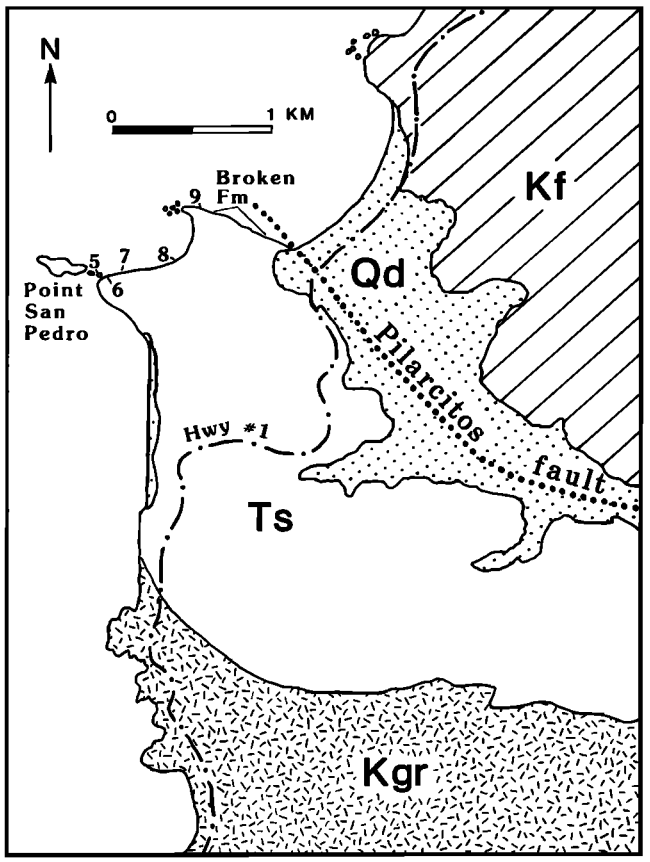


Fig. 12. General geology of the Point San Pedro area, showing location of sedimentary sequences from which paleomagnetic and/or paleocurrent information has been taken. Site numbers correspond to Table 2. Kf represents undifferentiated Franciscan rocks, Kgr marks granitic rocks of Montara Mountain, Ts is Paleocene strata of Point San Pedro, and Qd represents all Quaternary deposits undifferentiated.

while in the southern block they are 135°. If we rotate the magnetic declinations from the northern and southern blocks to a common value (Cretaceous north), the disparity in these current directions greatly diminishes (Figure 10). Current directions from turbidite layers become 280° and 250° and conglomeratic imbrications become 150° and 170° for the northern and southern blocks, respectively. Previously, this discrepancy in paleocurrent directions had been attributed to differences in source regions [*Howell and Vedder*, 1978]. The present analysis suggests that a better explanation involves tectonic rotation of the strata around axes perpendicular to the bedding planes that affected both magnetic declinations and paleocurrent directions by the same amounts.

The structurally corrected magnetic inclination values from the four stratigraphic sequences agree well with each other. The inclination values are all much shallower than the expected Cretaceous inclination value of 66° for the sampling locations. Calculation of the direction of magnetization for the samples from the Pigeon Point Formation is somewhat complicated. Simple vector averaging would lead to an incorrect mean direction of magnetization due to the between site declination differences. Instead, we have only considered inclination values for averaging and have elected to use the method proposed by *McFadden and Reid* [1982]. We have also elected to give each core unit weight rather than each site to allow for smaller numbers of cores from some locations, particularly, Pebble Beach and Franklin Point. The calculation was done by first converting each inclination to a virtual geomagnetic latitude (VGL). For the 118 cores from the Pigeon Point strata that were cleaned of VRM the mean VGL from *McFadden and Reid*'s [1982] method is 21.2° ($k = 29.3$, $\alpha(95) = 2.0$). The $\alpha(95)$ value calculated in this way is too small because every core does not have an independent structural attitude. A better figure for a 95% confidence level can be calculated for the $n = 24$ independent structural attitudes taken, giving a value of $\alpha(95) = 5.3°$. This paleolatitude corresponds to a position adjacent to the present location of the mid-Americas trench off southwestern Mexico (Figure 11).

Geology of Point San Pedro

In addition to the locations in the Pigeon Point Formation, paleomagnetic samples were also taken from Paleocene rocks at Point San Pedro, San Mateo County, California (Figure 12). The rocks lie in buttress unconformity on the Montara Granodiorite and are structurally juxtaposed across the Pilarcitos fault to the Franciscan assemblage. These strata are at least 1000 m thick and represent inner to middle fan environments of a submarine fan complex [*Howell et al.*, 1977; *Morgan*, 1981]. The age of the sedimentary rocks at Point San Pedro has been an object of some confusion. They originally were regarded as a basal member of the Franciscan Formation by *Lawson* [1895], but later *Darrow* [1963] found the strata contained early Tertiary mollusks and divided it into a Late Cretaceous lower part and Paleocene upper part. Most recent age assignments, from clams and gastropods in particular, are Paleocene (Ynezian) for the entire sequence [*Morgan*, 1981].

The rocks at Point San Pedro are extensively faulted and folded. The strata are divisible into three fault-bounded blocks of uncertain lateral correlation [*Morgan*, 1981]. In the Shelter Cove area, where we sampled, the exposure is good, and the structure is complex and consists of alternating sequences of upright and overturned thin-bedded turbidites, folded around subhorizontal axes. The age of folding is not known other than that it predates the deposition of the overlying early Miocene Monterey Formation.

Procedures

The basic experimental plan for the magnetic work on the samples from Point San Pedro closely paralleled that for the samples from Pigeon Point. Approximately, 80 cores were drilled from sandstone layers in Shelter Cove. The range of NRM intensity was from 4×10^{-7} to 3×10^{-6} emu/cm^3, consistent with NRM intensities of the Pigeon Point samples, but an equal-area plot of the distribution of NRM directions revealed a chaotic pattern, very unlike the samples from Pigeon Point. We were initially suspicious that the magnetizations might be biased toward the axes of the drill cores but found instead that the directions of magnetization tended to be perpendicular to the core axes. Close inspection of the specimens with the strongest magnetizations found a distinct metallic patina on the ends of the cores which presumably came from the diamond saw blades used to cut the cores. Wet grinding of the ends of the specimens to remove this metallic patina resulted in a drastic change in magnetic direction and decrease in intensity of magnetization an average of sevenfold.

While the grinding effectively removed the magnetic effect of the metalic patina, the low intensities of magnetization remaining required extreme caution during subsequent magnetic cleaning procedures. Progressive af demagnetizations at 25-Oe

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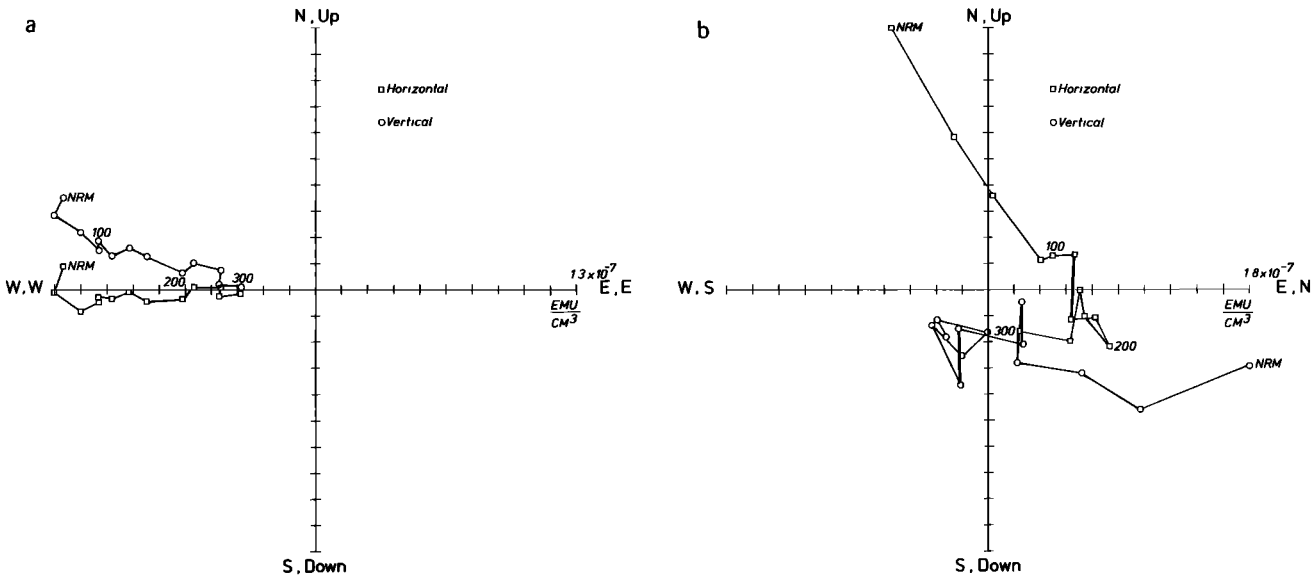


Fig. 13. Orthogonal vector component diagrams of alternating field demagnetization of samples from the Paleocene strata of Point San Pedro. (*a*) Sample from site 8 showing cleaning producing a linear descent to the origin. (*b*) Sample from site 7 showing low coercivity, weak magnetization, and erratic cleaning behavior.

intervals were performed on one specimen from each core to a peak field strength of 300 Oe. This field strength was sufficient to remove secondary components of magnetization and isolate the primary magnetization. Coarse-grained sandstones generally had less stable magnetizations, so that af demagnetization frequently did not result in isolation of primary directions of magnetization. A second set of 80 cores was collected from fine-grained, thin-bedded turbidites (Bouma T_{dc} intervals) and wet ground to remove possible metallic contamination. The finer-grained specimens, although as weakly magnetized as the sandstones, were much more stable in response to af demagnetization and almost always gave consistent cleaned directions of magnetization.

Paleomagnetic Results From Point San Pedro

A total of 156 cores were drilled in the Shelter Cove area of Point San Pedro at approximately 1-m stratigraphic intervals. Sampling was done at four separate outcrops (Figure 12), including one on a nearby offshore islet, spanning a total east-west separation of 350 m. The beds at all four outcrops strike near 110°, but the dips vary from outcrop to outcrop from approximately 50° for an entirely upright section (site 8) to 65° for two overturned sections (sites 5 and 7) and from slightly overturned (85°) to upright (60°) within another sampling cluster (site 6).

After completion of the stepwise af cleaning on each specimen the results from 102 specimens cleaned of secondary magnetic components were averaged together by site. Of these, 38 samples have vector component diagrams with linear segments that pass through the origin for some portion of their demagnetization behavior (Figure 13*a*). The remaining 64 samples show good magnetic cleaning and stable magnetic directions but do not produce linear segments on vector component diagrams. Three quarters (40 of 54) of the samples rejected were medium-grained sandstone, Bouma T_a or T_{bc} units (*Bouma*, 1962]. Samples were rejected principally because of unstable cleaned directions of magnetization due to low coercivities and excessive VRM components directed along the present field. Most of the rest of the specimens rejected (10 of 54) had visible evidence of authigenic ferric oxide minerals (Figure 13*b*). Both of these negative attributes are associated with coarseness of the grain size of the magnetic carrier and were minimal in the finer-grained samples.

Equal-area plots of the average directions of magnetization before structural correction show both positive and negative inclination values, but correcting by rotation around the strike of bedding produces consistent inclination values near −40° (Figure 14 and Table 2). Declinations, however, were initially distributed toward the southwest but disperse into an arc with values from 234° to 320°. The inclination values are all considerably shallower than the −63.5° value expected for a reversed geomagnetic field at the location of Point San Pedro in Paleocene time. Declination values are up to 150° clockwise from the expected Paleocene reversed declination value of 169°.

Fold tests were performed on the paleomagnetic results from several pairs of the outcrops, before and after structural correction. The diversity of dip values would be expected to make the fold tests quite conclusive, despite a common value for the strike of the beds. Outcrops which we compared were the pairs 7 against 8, 6A against 6B, and 6B against 7 using the method of *McFadden and Jones* [1981]. All three pairs of outcrops were found to be distinct from one another at the 95% confidence level before unfolding. Unfortunately, this generally remained the case after structural correction as well, with the exception of outcrop pair 6B and 7, which became indistinguishable at the 95% level. When the same test is done using only inclinations, the result is positive at 95% confidence for all three pairs. The pairs 5 and 6A were not tested because their structural attitudes are similar.

The smear of declination values at Point San Pedro is not random but appears to be related to proximity to the Pilarcitos fault. The Pilarcitos fault passes only 700 m northeast of the easternmost outcrops at Shelter Cove (Figure 12). The upright outcrop farthest to the east and nearest the fault has been rotated clockwise more than the other outcrops. The slightly overturned outcrop at the west end of the beach exposures has been rotated the least. The Pilarcitos fault is an ancestral strand of the San Andreas fault and has at least 275 km of Neogene right slip. Given the close proximity of the

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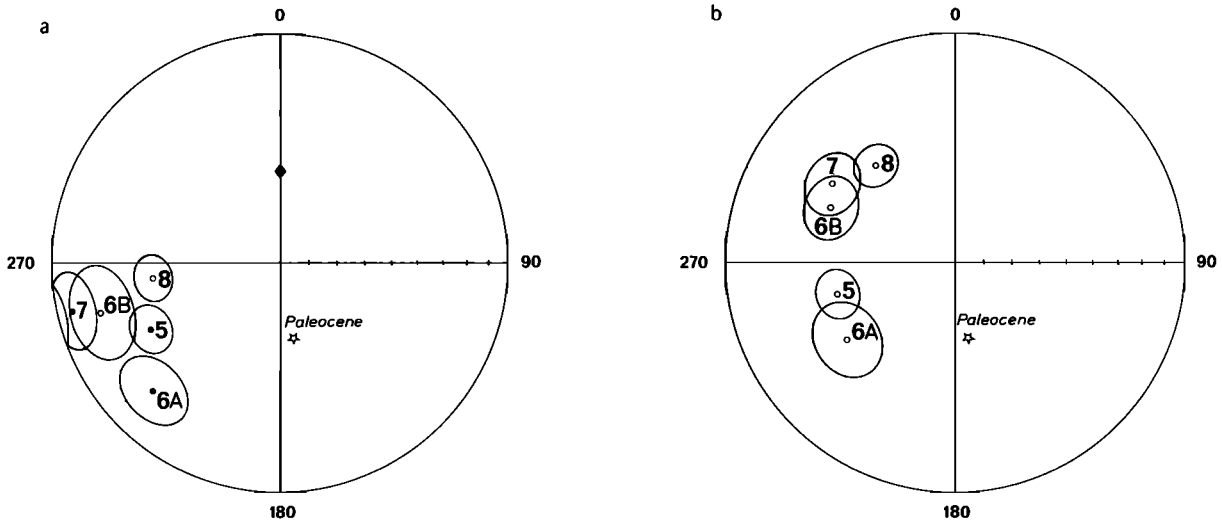


Fig. 14. (*a*) Equal-area plot of structurally uncorrected af-cleaned field directions with 95% confidence circles from four outcrops in the Paleocene rocks at Point San Pedro. Open star represents expected Paleocene reversed field direction for Point San Pedro. The diamond shows the present axial dipole direction for the collecting sites. (*b*) Equal-area plot of structurally corrected af-cleaned field directions from four outcrops in the Paleocene rocks at Point San Pedro.

easternmost sampled outcrops at Shelter Cove to the Pilarcitos fault, it is reasonable to expect large differences in the amount of apparent clockwise rotation as one moves westward along the beach outcrops, even over a distance of only 350 m. Because the strike of the beds is uniform across this transect of varying rotations, the rotations must have occurred prior to folding and would have involved faulting along bedding planes.

For an independent check of the rotation hypothesis we made a detailed analysis of paleocurrent directions across the transect of rotated declinations. Because the apparent magnetic declinations are so disparate, we were expecting to find a consistent correlation of the declinations with flow direction indicators despite the azimuthal uncertainties of sedimentary structures (small-scale crossbeds, flutes, and grooves) as well as the dispersion of current patterns characteristic of a submarine fan environment. Figure 15 shows the results of this experiment, treating each paleocurrent measurement as a horizontal unit vector and using a statistical method proposed by *Curray* [1956] to calculate the standard deviation. There is a tendency for the flow directions in the more easterly outcropping beds to be rotated in a clockwise sense relative to those in the western outcrops, although the spread is not as large as the paleomagnetic data would suggest and could have been caused solely by normal shifts in distributary channels. What is of more interest, however, is the overall north to northwest trending flow direction. Holocene to recent submarine fans in open basins along the west margin of North America consistently show patterns of flow to the west, southwest, and south. These patterns result from the Coriolis effect which causes the northern side levee to build higher, forcing the channel to migrate and turn toward the south. Flow data from most ancient submarine fans in coastal North America also indicate west to south dispersive patterns; two notable exceptions are north directed flow directions in Eocene strata of the Tyee Formation of Oregon and in Eocene strata on San Miguel Island of the California continental borderland. Yet in both instances, recent paleomagnetic studies indicate 70° of postde-

TABLE 2. Paleomagnetic Data After af Demagnetization, Both Structurally Uncorrected and Corrected, and Paleocurrent Data ($\pm 1\sigma$) Taken From Five Outcrops in the Shelter Cover Area at Point San Pedro

Site	Data	*N*	*R*	*k*	$\alpha(95)$	*I*	*D*	Current Direction
5	uncorr		24.05951	12.9	8.2	36.3	242.5	311 ± 17
	corr	26	24.11633	13.3	8.1	−45.7	254.0	(15)
6A	uncorr		12.04674	12.6	12.2	22.0	224.8	322 ± 25
	corr	13	11.99323	11.9	12.5	−41.3	233.8	(16)
6B	uncorr		6.69471	19.7	14.0	−19.4	254.2	
	corr	7	6.80625	31.0	11.0	−40.8	292.9	
7	uncorr		19.09560	10.5	10.3	7.0	256.7	349 ± 19
	corr	21	18.90184	9.5	10.9	−37.0	302.0	(23)
8	uncorr		32.05603	11.5	7.5	−43.1	262.7	336 ± 21
	corr	35	31.93739	11.1	7.6	−45.1	320.1	(22)
9								4 ± 32
	...	...	...	...	...	...	...	(18)

Current directions measured in degrees east of north; numbers in parenthesis beneath current directions are total number of current indicators found for each site. Site numbers correspond with those on Figures 12 and 14.

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positional clockwise rotation [*Simpson and Cox*, 1977; *Kamerling and Luyendyk*, 1981]. Thus the overall northwestward flow direction indicated for the Point San Pedro strata is suspect, and the clockwise rotation of at least 90° indicated by the paleomagnetic declinations removes the anomaly in the paleocurrent directions.

The paleomagnetic results from the four outcrop areas in Shelter Cove at Point San Pedro have been averaged in the same way as were the Pigeon Point data, with the addition of a second (overall) fold test. For the 102 cores that were considered to have been completely cleaned magnetically, the inclinations of magnetization were converted to two sets of VGL's: one set for the in situ directions and one set after structural correction. The method of *McFadden and Reid* [1982] was used to calculate mean paleolatitudes and associated statistical parameters. The precision parameter k increased from 7 to 25 after structural correction, and because the beds at Point San Pedro vary as much as 70° in dip, this is a conclusively positive result. The mean paleolatitude is 24.5° with a 95% confidence interval $\alpha(95)$ of 2.4°. A better estimate of $\alpha(95)$ is 4.0°, calculated from the same precision parameter but for $n = 53$ independent measurements of structural attitude. This paleolatitude corresponds to the present location of the mouth of the Gulf of California (Figure 16).

Northward Motion of the Salinian Terrane

Calculation of paleolatitudes from the mean inclination values found for the Pigeon Point Formation and the Paleocene strata at Point San Pedro allows us to estimate the position of the northern part of the Salinian terrane against age. Because geologic relations suggest the Salinian terrane was part of the agglomerated Santa Lucia–Orocopia allochthon by latest Cretaceous time, by inference we can estimate paleolatitudinal travel paths for most of southwestern California.

Expected latitude positions for the Salinian terrane can be determined from paleomagnetic poles of different ages from geologic units located on cratonic North America. For this purpose we use a reference position of 37.5°N, 237.5°E in central San Mateo County, California, located midway between Pigeon Point and Point San Pedro.

Latitude positions for the Salinian terrane can also be determined from the known offsets along the San Andreas and San Gregorio faults that occurred primarily in Neogene time. This method of calculation tacitly assumes that no major latitudinal changes have occurred in westernmost North America east of the San Andreas fault since Cretaceous time, an assumption

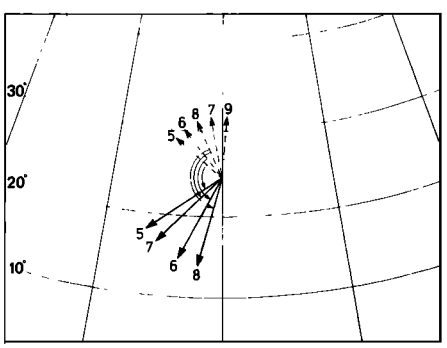


Fig. 15. Paleocurrent data from five outcrops at Point San Pedro displayed as measured (dashed) and after normalization to 180° declination (solid) at their probable latitude of origin in Paleocene time. Numbers relate to outcrop identifiers on Table 2 and Figure 12.

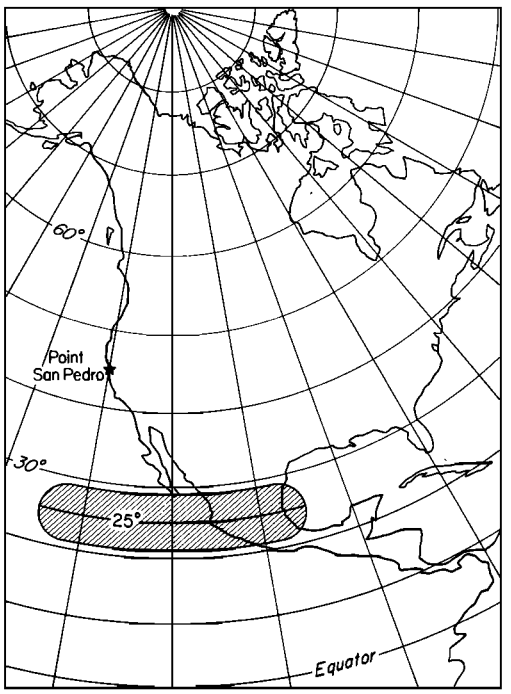


Fig. 16. Equal-area map of North America showing suggested latitude band of origin for Paleocene rocks at Point San Pedro. Latitude-longitude grid corresponds to Paleocene paleomagnetic pole position (see Table 3).

supported by the cratonic character of paleomagnetic data from Jurassic plutons in the Sierra Nevada [*Gromme et al.*, 1967]. Estimation of displacements on the San Andreas fault is based on correlation of molluscan faunal patterns [*Addicott*, 1968] for early, middle, and late Miocene time and on correlation of the stratigraphic similarities of the Pinnacles and Neenach volcanic areas [*Matthew*, 1973] and of the San Juan Bautista and San Emigdio volcanic areas [*Huffman*, 1972]. The correlations of early Miocene deposits suggest that approximately 305 km of right-lateral displacement has occurred on the San Andeeas fault since that time. Important correlations have also been made between early Eocene submarine fan deposits of the Butano Sandstone in the Santa Cruz Mountains and the Point of Rocks Sandstone in the northern Temblor Range [*Nilsen and Clarke*, 1975]. These sandstone bodies appear to be offset about 305 km, despite a faunal age of Ulatisian-Narizian (51–47 m.y.), 25 m.y. older than the Miocene units offset the same amount. This condition indicates a lack of movement on the San Andreas fault between Eocene and Miocene time.

Deformation on the San Gregorio fault also needs to be considered in any geologic reconstruction of the paleolatitude of the Pigeon Point Formation. This fault has been active during Neogene time, and even in late Pleistocene time has had active right-lateral displacement approaching 2 cm/yr [*Howell et al.*, 1980; *Weber and Lajoie*, 1979]. Estimates of total displacement vary to some extent but suggest at least 90 km of right-lateral motion since Miocene time [*Mullins and Nagel*, 1981; *Graham and Dickinson*, 1978].

The various estimates of the paleolatitude of the Salinian terrane predicted on one hand from paleomagnetic poles from cratonic North America and on the other hand from geologic offsets can be compared with the paleomagnetic evidence derived from the Pigeon Point Formation and the strata at

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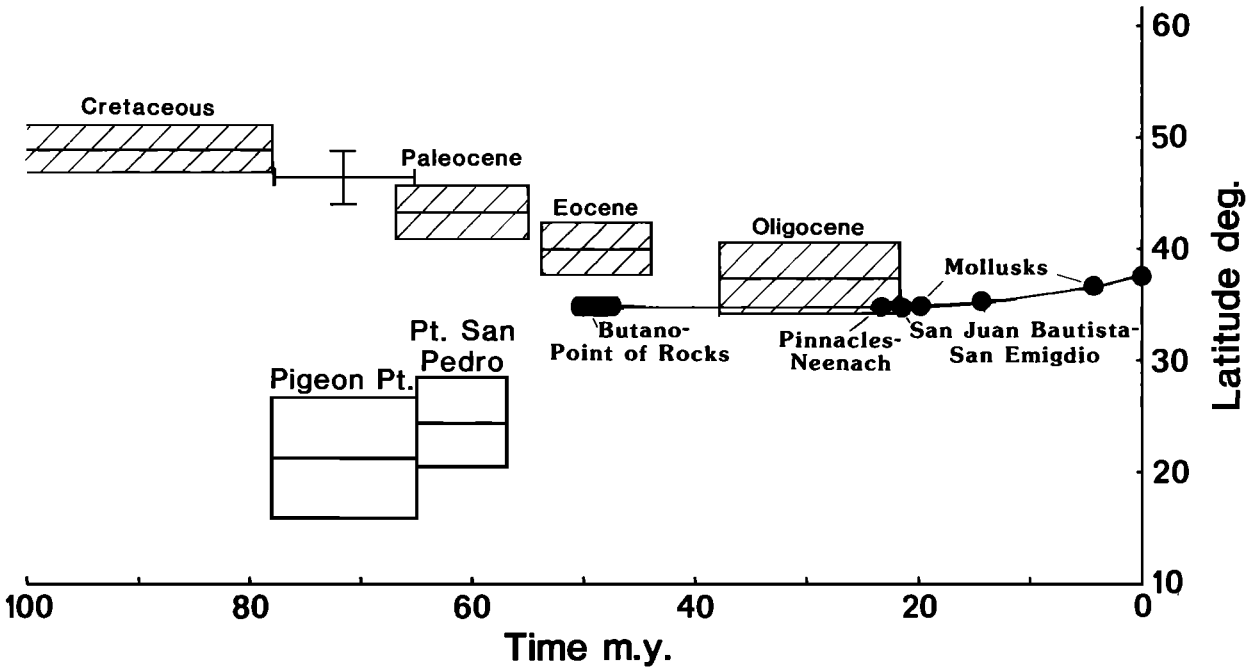


Fig. 17. Diagram of predicted and observed paleolatitudes against time for the Salinian terrane. Diagonal ruled boxes represent predictions derived from paleomagnetic data for cratonic North America (Table 3). Cross is prediction for Pigeon Point Formation obtained by interpolation between Cretaceous and Paleocene data. Open boxes are observed data from Pigeon Point Formation and rocks at Point San Pedro. Vertical dimensions of boxes are 95% confidence ranges, and horizontal dimensions are published or estimated age ranges. Solid symbols represent estimates from reconstruction of known offsets of geologic units separated by major right-lateral faults (see text.)

Point San Pedro (Figure 17 and Table 3). What is immediately evident is that both stratigraphic sequences were located far south of the predicted paleolatitudes for either Cretaceous or Paleocene time assuming that there has been no motion of the Salinian terrane relative to cratonic North America. For the Pigeon Point Formation the difference is as much as 25°–31° of latitude. These results also are as much as 20° lower in latitude when compared to the paleolatitude position reconstructed from offsets of younger geologic units. An early Tertiary proto San Andreas right-lateral motion of the Salinian terrane of the order of 1500 km northward is required to account for the paleomagnetic data. This estimate of proto San Andreas motion is very much larger than earlier figures of 220–420 km required to juxtapose geologically correlative units or features [*Nilsen and Clarke*, 1975].

An important aspect of the paleolatitude estimates shown on Figure 17 comes from the predicted paleolatitude variation for the reference position in San Mateo County. What is indicated is an apparent decrease in latitude for the western edge of North America, particularly in the interval of time between Late Cretaceous and Eocene. The apparent polar wander path for North America when viewed from the midcontinent appears as a counterclockwise plate rotation at nearly constant colatitude; however, at the western edge of North America this rotation produces motion to the south of the order of 500–1000 km. Thus a northward moving Salinian terrane and a southward moving west coast of North America may have been converging toward one another in Late Cretaceous–early Tertiary time.

Rates of motion, both absolute and relative, can be calculated from the paleolatitude estimates on Figure 17. An overriding assumption is that the western coastline of North America was essentially north-south so that longitudinal motion was small and unimportant. An absolute rate of 8.5 cm/yr is required to produce 14° of northward latitude motion for the Salinian terrane during a time interval of 20 m.y. (Late Cretaceous to the early Tertiary docking with North America); this rate is 6.5 cm/yr if 25 m.y. is taken as the time interval. The 9° of southward motion of the west coast of North America (Figure 17) is better constrained to an interval of 30 m.y., corresponding to a rate of about 3 cm/yr of absolute motion. Together, these absolute rates suggest relative rates of motion between 10 and 12 cm/yr between North America and the Salinian terrane during early Tertiary time. While high, these rates are within the range of motions measured for present plate interactions [*Minster and Jordan*, 1978].

Northward Shift of the San Simeon, Stanley Mountain, and Tujunga Terranes

Preliminary paleomagnetic results from the Cambria Slab of the San Simeon terrane suggest paleolatitudes of approximately 20°–25° for sedimentary rocks of Late Cretaceous age (D. E. Champion, unpublished data, 1982). *McWilliams and Howell* [1982] report anomalously low paleomagnetic inclination data from Upper Cretaceous Cenomanian-Santonian rocks in the San Rafael Mountains (southeast part of Stanley Mountain terrane). This evidence lends credence to both an Upper Cretaceous suturing of the Salinian and Sur-Obispo terranes [*Vedder et al.*, 1983] as well as a more southerly origin for all of these terranes.

Preliminary paleomagnetic results have also been obtained from the Upper Cretaceous to lower Tertiary San Francisquito Formation located in the San Gabriel Mountains of southern California (D. E. Champion, unpublished data, 1981). These strata lie on one of the many fault-bounded

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TABLE 3. Predicted Paleolatitude and Paleomagnetic Field Directions at Reference Locality (37.5°N, 237.5°E)

Time	L	I	D
Oligocene (22–38 m.y.)	37.3 ± 3.2	56.7 ± 3.1	351.5 ± 4.0
Eocene (44–54 m.y.)	40.0 ± 2.3	59.2 ± 2.1	351.3 ± 3.1
Paleocene (55–67 m.y.)	43.3 ± 2.5	62.0 ± 2.1	351.8 ± 3.4
Cretaceous-Paleocene*	46.5 ± 2.3	64.6 ± 1.8	343.0 ± 3.4
Cretaceous (78–135 m.y.)	48.9 ± 2.2	66.5 ± 1.6	333.3 ± 3.3

L, latitude in degrees north. I, inclination in degrees downward. D, declination in degrees eastward. Uncertainties are 95% confidence intervals calculated by the method of *Beck* [1980], as modified by *Demarest* [1983]. Reference poles for Tertiary time are from *Diehl et al.* [1983]. Reference pole for Cretaceous time is mean of entries 39, 40, 42, 43, 44, and 45 of Table I of *Irving and Irving* [1982].

*Calculated from average of Paleocene and Cretaceous poles.

blocks of the Tujunga terrane. The results again suggest paleolatitudes of between 20° and 25° and clockwise rotations of about 90°. The rotation corresponds well with paleomagnetic results from Miocene volcanic rocks in the Transverse Ranges, which also indicate 90° of clockwise rotation since Miocene time [*Luyendyk et al.*, 1980].

CONCLUSIONS

The paleomagnetic experiments indicate that Upper Cretaceous and Paleocene sedimentary strata of the Salinian terrane were deposited 2800 and 2100 km, respectively, south of their present locations with uncertainties of the order of 500 km. Basin analysis studies suggest that during the Late Cretaceous and Paleocene episode of deposition the Salinian terrane was in a wrench tectonic mode [*Howell and Vedder*, 1978]. Additionally, the Upper Cretaceous fluviodeltaic sequence in the southeastern Stanley Mountain terrane provides evidence that these two terranes had become attached by Campanian time (70–75 m.y. B.P.) and that they traveled side by side during the episode of northward drift. The abundance of siliceous volcanic clasts in many of the Upper Cretaceous and lower Tertiary sequences of these terranes indicate contiguity with the western cordillera at some as yet unidentified place.

On the basis of geologic correlations and preliminary paleomagnetic results, other terranes shared in this northward translation; these include rocks in the San Gabriel, Orocopia, and Chocolate mountains [*Crowell*, 1962; *Ross*, 1977; *Vedder et al.*, 1983]. This amalgamated terrane, involving much of the southern Coast Ranges and southwestern Mojave Desert, is here called the Santa Lucia–Orocopia allochthon and includes the "single allochthon" of *Haxel and Dillon* [1978], part of which corresponds to the upper plate of the Vincent-Orocopia-Chocolate mountains thrust system. The Baldy terrane also may be incorporated in the allochthon, though the origins of the Pelona, Orocopia, and Rand schists are uncertain. Alternatively, the Baldy terrane may represent ocean margin deposits of pre-Tertiary age that were overridden during collision of the Santa Lucia–Orocopia allochthon with North America.

Neogene transcurrent faulting in western California has generated as much as 1000 km of cumulative right slip [*Atwater*, 1970]. The paleomagnetic data require additionally at least 1500 km of pre-55-m.y. relative motion in the same sense. Rocks used for the paleomagnetic measurements are late Campanian and/or early Maestrichtian and Paleocene, covering an age span from about 75 to 60 m.y. B.P. A hypothetical continental transcurrent fault in the present vicinity of Central America could have resulted in some of the additional relative motion. From 60 to 55 m.y., the approximate age span of the erosional unconformity throughout the allochthon [*Vedder et al.*, 1983], relative northward transport may have occurred without physical connection with the cordillera to the east.

The allochthon was accreted to the southern California part of North America before Eocene time (pre-55 m.y. B.P.) because lower Eocene rocks lap across the allochthon and onto stable California terranes, as indicated by reconstruction of the Butano–Point of Rocks fan system of *Nilsen and Clarke* [1975]. Figure 17 records the slip history required by the paleomagnetic results; these data imply an average northward plate velocity of 8.5 cm/yr for the Salinian terrane.

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